

Conformal prediction or the good old DKW from the 50s?

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Editor: Ernst Ahlberg, Ulf Johansson, Henrik Boström, Alberto Carlevaro, Johan Hallberg Szabadváry and Lars Carlsson

Abstract

Prediction sets based on Conformal inference, the Dvoretzky–Kiefer–Wolfowitz–Massart (DKW) inequality, and classical tolerance-region theory are often discussed as if they are competing for the same job while they are not. Each approach controls a different random quantity. In this paper, we organize the three tools around a single construction: pick a map whose image distribution is known, choose a high-mass subset on that scale, then *pull back*. CP, DKW and the Beta law of Wilks (1941) differ only in which map and which mass measure they use. However, putting these methods on the empirical CDF-scale puts all three on a common axis. With this, two simple inclusions appear. Ordinary CP intervals sit inside DKW intervals at the same nominal confidence level, but the two procedures deliver different guarantees. When the conformal procedure is recalibrated through the Beta law to deliver the same training-conditional (PAC) guarantee as DKW, the conformal interval is still contained in the DKW one. DKW controls the whole empirical path, whereas CP with the Beta law controls one chosen order-statistic interval. We trace the line from the industrial quality control of Shewhart (1931) through the finite-sample tolerance theory of Wilks (1941) to the conformal formulation of Gammernan et al. (1998) and Vovk et al. (2005).

Keywords: Conformal prediction; DKW inequality; tolerance regions; Beta distribution; distribution-free uncertainty quantification; empirical process.

1. Introduction

We observe real-valued data $D_n = (Z_1, \dots, Z_n)$ and aim to predict where the next observation Z_{n+1} will land. For simplicity, we assume that $Z_1, \dots, Z_{n+1}$ are *iid* from a distribution with continuous CDF F . The goal is to construct a set $\hat{Q}$ that contains Z_{n+1} with probability at least $1 - \alpha$. A useful recipe is to transform the prediction problem to a scale on which probabilities are known. Let $\Pi : \mathbb{R} \rightarrow \mathcal{Y}$ be a map such that $\Pi(Z_{n+1})$ has known distribution μ . If $A \subseteq \mathcal{Y}$ satisfies $\mu(A) \geq 1 - \alpha$, then its pullback

$$Q_\Pi := \Pi^{-1}(A) = \{z : \Pi(z) \in A\}, \quad \mathbb{P}(Z_{n+1} \in Q_\Pi) = \mu(A) \geq 1 - \alpha.$$

If the ground-truth distribution F were known, it would suffice to take $\Pi = F$ and the probability integral transform gives $F(Z_{n+1})$ is uniform on $(0, 1)$. Pulling back the central interval $A_\alpha = [\alpha/2, 1 - \alpha/2]$ gives the oracle prediction set which has coverage exactly $1 - \alpha$.

$$Q_{1-\alpha}^{\text{oracle}} = F^{-1}(A_\alpha) = [F^{-1}(\alpha/2), F^{-1}(1 - \alpha/2)],$$

When $\hat{\Pi}$ is used in place of Π , there are two ways to retain calibration. If the finite-sample law of $\hat{\Pi}(Z_{n+1})$ is known, we apply the pullback construction using that law. Otherwise, we identify a high-probability event on which a set constructed from $\hat{\Pi}$ contains the oracle set. Such an event may control the approximation uniformly, as in DKW, or only at the target boundaries, through pointwise concentration or exact finite-sample laws. In this paper, $\Pi = F$ and $\hat{\Pi} = F_n$, where $F_n(t) = n^{-1} \sum_{i=1}^n \mathbb{1}\{Z_i \leq t\}$ denotes the empirical CDF. CP (Gammerman et al., 1998) and DKW use the same empirical CDF in two different ways:

$$\begin{aligned} \text{Conformal prediction: } F_n(Z_{n+1}) &\sim \text{Unif}\{0, 1/n, \dots, 1\}, \\ \text{DKW: } \mathbb{P}(\|F_n - F\|_\infty \leq \varepsilon_n(\delta)) &\geq 1 - \delta. \end{aligned}$$

The first statement follows from the uniform rank of Z_{n+1} under exchangeability, and the second treats F_n as an estimator of F and gives a high-probability inflation of the oracle set. These two uses of F_n come from different historical traditions. Classical tolerance theory provides the bridge between them: Wilks (1941) derived the exact Beta law of the conditional coverage of order-statistic intervals. This law allows conformal prediction to be calibrated to the same training-conditional, or PAC, target as DKW. Recent expository writing has placed conformal prediction in tension with DKW (Recht, 2024) and essentially argues that DKW should be preferred to CP for distribution-free prediction. The comparison is delicate because the two methods control different random quantities and provide de facto different guarantees. Our aim is to align those guarantees, place the resulting prediction sets on a common scale, and compare them precisely. The blog post has some interesting provocative insights, though. The reasoning is that ordinary CP gives marginal coverage, which averages over the random calibration sample, whereas a practitioner with one fixed calibration sample may want a training-conditional or PAC guarantee: with high probability over the calibration sample, the realized coverage is at least $1 - \alpha$. DKW delivers this kind of guarantee directly; ordinary CP does not. Recht then goes further, writing that “the worst part is I think they are quantifying uncertainty that no one actually cares about” (Recht, 2024). There he is referring to object-conditional coverage, which is distinct from the PAC target and cannot be obtained distribution-free without additional assumptions (Lei and Wasserman, 2014; Foygel Barber et al., 2021). The post concludes that one should therefore use the older DKW inequality instead. This conclusion does not follow. CP can be calibrated through Wilks’s Beta law for realized coverage to deliver the same PAC guarantee as DKW (Vovk, 2013; Hulsman, 2022). When this calibration is done, the resulting CP set is contained in the DKW set at the same target, in nondegenerate regimes. The reality is that there is no need to fight: all these prediction sets are more or less constructed from the same empirical CDF, and they differ only in the random quantity they control. DKW is broader in scope (it controls $F_n - F$ uniformly over all thresholds, not just one), and that broader scope costs the $1/\sqrt{n}$ rate. At a single fixed target, CP with Beta calibration is sharper than DKW. At many simultaneous targets, DKW remains the right tool. This will become precise in section 4. Wilks’s exact coverage law has been re-expressed in conformal language by Vovk (2013), Hulsman (2022), and Marques F. (2025); the last of these studies the finite- and large-batch formulations recalled in Appendix C.

2. Historical arc

Quality control (1931): prediction limits for the next observation. [Shewhart \(1931\)](#) faced a practical problem in industrial statistics. A factory produces parts whose dimensions vary (following some unknown distribution), and the foreman wants to know whether the production process is operating consistently. From past measurements, we compute the mean $\bar{Z}$ and the standard deviation s . Draw two horizontal lines at $\bar{Z} - 3s$ and $\bar{Z} + 3s$, the *three- σ control limits*. Plot each new measurement on this chart. As long as new measurements fall between the limits, the process is consistent with what came before. When a measurement falls outside, something has changed and the foreman investigates. This is, in modern vocabulary, a prediction interval for the next observation, built from past observations, used to flag when prediction fails. The question Shewhart asked is exactly the one conformal prediction would address seventy years later. Past data, new point, prescribed probability of the new point falling in a chosen region.

The empirical CDF and its rate (1933–1990). The natural substitute for F is the empirical CDF F_n . The first question is whether this substitution is consistent. [Glivenko \(1933\)](#) and [Cantelli \(1933\)](#) answered it affirmatively.

Theorem 1 (Glivenko–Cantelli) *Suppose $Z_1, \dots, Z_n$ are iid with continuous CDF F . Let F_n denote their empirical CDF. Then,*

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \xrightarrow{\text{a.s.}} 0.$$

Glivenko–Cantelli establishes uniform consistency, but it gives no finite-sample error bound. Appendix A gives the elementary discretization proof following the lecture notes of [Tsybakov \(2007\)](#). The next question is the size of the error. The asymptotic answer is due to [Kolmogorov \(1933\)](#) and [Smirnov \(1944\)](#).

Theorem 2 (Kolmogorov–Smirnov limit) *In the setting of theorem 1, for every $\lambda > 0$,*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\sqrt{n} \sup_{t \in \mathbb{R}} |F_n(t) - F(t)| > \lambda \right) = 2 \sum_{k=1}^{\infty} (-1)^{k+1} e^{-2k^2 \lambda^2}.$$

The right-hand side is the tail probability of the largest absolute excursion of a Brownian bridge. Thus, the theorem identifies the $n^{-1/2}$ scale, but remains asymptotic. [Dvoretzky et al. \(1956\)](#) obtained a finite-sample exponential inequality with a universal constant C , and [Massart \(1990\)](#) proved that one may take $C = 2$ and that this value is optimal.

Theorem 3 (DKW–Massart inequality) *Let $Z_1, \dots, Z_n$ be iid with distribution function F , and let F_n be their empirical CDF. Then, for every $\varepsilon > 0$,*

$$\mathbb{P} \left(\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| > \varepsilon \right) \leq 2e^{-2n\varepsilon^2}.$$

Moreover, the constant 2 is optimal: no smaller universal constant makes the inequality valid for every n and ε .

The DKW inequality controls the entire empirical path. Two observations clarify the difference between this uniform control and what is sufficient at one fixed prediction level.

1. *Reduction to uniforms.*

If F is continuous and $U_i = F(Z_i)$, then $U_1, \dots, U_n$ are *iid* uniform on $[0, 1]$. Let G_n be their empirical CDF. If F is strictly increasing, then

$$\mathbb{1}\{Z_i \leq t\} = \mathbb{1}\{F(Z_i) \leq F(t)\}, \quad F_n(t) = G_n(F(t)).$$

Since a continuous CDF ranges through every value in $(0, 1)$, this gives

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| = \sup_{u \in [0, 1]} |G_n(u) - u|. \quad (1)$$

If F has flat portions, the indicator identity can fail only on level sets carrying zero probability, so eq. (1) still holds almost surely. This can also be seen directly from the order statistics. Let $Z_{(1)} \leq \dots \leq Z_{(n)}$ and $U_{(j)} = F(Z_{(j)})$. Continuity of F gives

$$\begin{aligned} \sup_t (F_n(t) - F(t)) &= \max_{1 \leq j \leq n} \left(\frac{j}{n} - U_{(j)} \right), \\ \sup_t (F(t) - F_n(t)) &= \max_{1 \leq j \leq n} \left(U_{(j)} - \frac{j-1}{n} \right). \end{aligned}$$

The same two formulas hold for $G_n(u) - u$, proving the equality for a general continuous CDF. At any fixed u , the variable $nG_n(u)$ has distribution $\text{Bin}(n, u)$, and Hoeffding's inequality gives $\mathbb{P}(G_n(u) - u > \varepsilon) \leq e^{-2n\varepsilon^2}$. For a uniform statement, the difficulty is the supremum over u : a naive union bound over the possible crossing points would lose a factor of order n , and Massart's proof avoids this loss. For one prespecified prediction level, however, the full supremum is unnecessary.

2. *A fixed target needs only two points.*

Let $a = \alpha/2$, $q_a = F^{-1}(a)$, and $q_{1-a} = F^{-1}(1-a)$. Pointwise Hoeffding bounds at the two oracle quantiles give

$$\mathbb{P}(F_n(q_a) < a - \varepsilon) \leq e^{-2n\varepsilon^2}, \quad \mathbb{P}(F_n(q_{1-a}) > 1 - a + \varepsilon) \leq e^{-2n\varepsilon^2}.$$

Taking a union bound and setting $\varepsilon = \sqrt{\log(2/\delta)/(2n)}$ makes both inequalities hold with probability at least $1 - \delta$. On this event, every integer k satisfying $k/n \leq a - \varepsilon$ obeys $Z_{(k)} \leq q_a$ and $Z_{(n-k+1)} \geq q_{1-a}$. Consequently, $[Z_{(k)}, Z_{(n-k+1)}] \supseteq [q_{\alpha/2}, q_{1-\alpha/2}]$, and its realized coverage is at least $1 - \alpha$. The oracle quantiles appear only in the proof; the resulting order-statistic interval is computable. Up to grid conventions, this gives the same cut as DKW,

$$\frac{k}{n} \approx \left(\frac{\alpha}{2} - \sqrt{\frac{\log(2/\delta)}{2n}} \right)_+.$$

Thus the size of the fixed- α DKW interval does not by itself result from uniform control: two pointwise Hoeffding bounds already certify essentially the same interval. The pointwise variables are exactly binomial, so Hoeffding can be sharpened. Indeed,

$$\mathbb{P}(Z_{(k)} > q_a) = \mathbb{P}(Z_{(n-k+1)} < q_{1-a}) = \mathbb{P}(\text{Bin}(n, a) \leq k-1).$$

Therefore the exact endpoint-wise calibration is

$$k_{\text{point}} = \max \left(\{0\} \cup \left\{ k \in \{1, \dots, \lfloor n/2 \rfloor\} : \mathbb{P}(\text{Bin}(n, \alpha/2) \leq k-1) \leq \frac{\delta}{2} \right\} \right).$$

The interval $[Z_{(k_{\text{point}})}, Z_{(n-k_{\text{point}}+1)}]$ then contains the oracle interval with probability at least $1 - \delta$. If a closed-form bound is preferred, the KL–Chernoff inequality gives, for $r < a$,

$$\mathbb{P}(F_n(q_a) \leq r) \leq \exp[-n \text{kl}(r \parallel a)].$$

Thus $n \text{kl}((k-1)/n \parallel \alpha/2) \geq \log(2/\delta)$ is a sufficient endpoint-wise condition, and is sharper than Hoeffding when α is small. Endpoint-wise calibration is still stronger than necessary: it requires each omitted tail to have mass at most $\alpha/2$. Wilks’s Beta law instead controls their combined mass. Writing $C_k = F(Z_{(n-k+1)}) - F(Z_{(k)})$ for the realized coverage of the central interval, Beta–binomial duality gives the exact PAC condition

$$\mathbb{P}(C_k < 1 - \alpha) = \mathbb{P}(\text{Bin}(n, \alpha) \leq 2k - 1) \leq \delta.$$

This directly targets realized coverage and is therefore generally sharper than separate control of the two endpoints.

These fixed-target PAC calibrations all have an $n^{-1/2}$ correction; they differ in sharpness and in whether they control the two tails separately or jointly. DKW additionally provides simultaneous validity over all thresholds and prediction levels. Appendix B gives the first-crossing argument behind Massart (1990)’s proof. The final calibration above relies on the classical tolerance-region law, to which we now turn.

Tolerance regions (1941–1968): the exact law of realized coverage. A different line of work, in exact finite-sample theory rather than uniform estimation, predates DKW. Classical authors developed distribution-free tolerance regions from order statistics (Wilks, 1941) (which, in our knowledge provides the first rigorous solution to Shewart’s problem and can be considered as first appearance of conformal prediction) and This was later generalized in (Wald, 1943; Scheffé and Tukey, 1945; Tukey, 1947; Fraser, 1951; Hill, 1968).

Wilks (1941) begins with a sample-size question: *“how large should our sample be in order for the tolerance range to have a given degree of stability?”* More precisely, for a given method of constructing tolerance limits L_1 and L_2 , he asks that the proportion of the population between them have average value a and lie between two prescribed values b and c with probability at least p .

Wilks denotes this random proportion by P . In our notation, it is the realized coverage

$$C := \mathbb{P}(Z_{n+1} \in [L_1, L_2] \mid \mathcal{D}_n) = F(L_2) - F(L_1).$$

Because L_1 and L_2 depend on the sample, C is itself a random variable. Wilks's question is therefore to determine the sample size for which this coverage has both an appropriate location and sufficient concentration:

$$\mathbb{E}[C] = a, \quad \mathbb{P}(b < C < c) \geq p.$$

Wilks constructs the limits by truncating the sample range, taking

$$L_1 = Z_{(k)}, \quad L_2 = Z_{(n-k+1)}.$$

His essential result is the exact finite-sample distribution of the realized coverage C .

Theorem 4 (Wilks's tolerance-interval law) *Let $Z_1, \dots, Z_n, Z_{n+1}$ be iid with continuous distribution function F , and let $Z_{(1)} \leq \dots \leq Z_{(n)}$ be the order statistics of the first n observations. Set $Z_{(0)} = -\infty$ and $Z_{(n+1)} = +\infty$. For $0 \leq r < s \leq n+1$, except $(r, s) = (0, n+1)$, the realized coverage of the order-statistic interval satisfies*

$$C_{r,s} := \mathbb{P}(Z_{n+1} \in [Z_{(r)}, Z_{(s)}] \mid \mathcal{D}_n) = F(Z_{(s)}) - F(Z_{(r)}) \sim \text{Beta}(s-r, n-s+r+1).$$

In particular, for $1 \leq k \leq \lfloor n/2 \rfloor$,

$$C_k := F(Z_{(n-k+1)}) - F(Z_{(k)}) \sim \text{Beta}(n+1-2k, 2k). \quad (2)$$

For $k=0$, the interval is the whole line and $C_0 = 1$ almost surely.

Proof Let $U_i = F(Z_i)$. By the probability integral transform, $U_1, \dots, U_n$ are iid uniform on $[0, 1]$, and $F(Z_{(j)}) = U_{(j)}$. Set $U_{(0)} = 0$ and $U_{(n+1)} = 1$. The uniform spacings satisfy

$$(U_{(1)} - U_{(0)}, \dots, U_{(n+1)} - U_{(n)}) \sim \text{Dirichlet}(1, \dots, 1).$$

Now $C_{r,s} = U_{(s)} - U_{(r)}$ is the sum of $s-r$ consecutive spacings. The aggregation property of the Dirichlet distribution therefore gives $C_{r,s} \sim \text{Beta}(s-r, n-s+r+1)$. $\blacksquare$

Wilks's law also gives the Beta-binomial distribution for the number covered in a finite future batch and its large-batch limit; we record this extension in Appendix C.

Wilks obtained the Beta law directly from multinomial counts over disjoint intervals; the Dirichlet spacing identity packages that calculation into a standard modern result. For a realized sample, C_k is the population mass contained in the resulting order-statistic interval. Across repeated samples, C_k is itself random. Wilks distinguished its mean, $\mathbb{E}[C_k] = a$, from its stability probability, $\mathbb{P}(b < C_k < c) \geq p$. For the central interval,

$$\mathbb{E}[C_k] = \frac{n+1-2k}{n+1}.$$

Thus, if the desired mean coverage is a , exact equality requires $2k = (1-a)(n+1)$. When this is not an integer, the largest rank with mean coverage at least a is $k = \lfloor (1-a)(n+1)/2 \rfloor$. The modern one-sided PAC criterion $\mathbb{P}(C_k \geq 1-\alpha) \geq 1-\delta$ is a lower-tail special case of the same stability calculation. Wilks also compared the nonparametric interval with the parametric Gaussian alternative $\bar{Z} \pm cs$. We return to these consequences in section 3.

Conformal prediction (1998): exact rank uniformity. A different route exploits finite-sample symmetry rather than controlling the discrepancy between F_n and F . Its key result characterizes the distribution of F_n at the future observation.

Lemma 5 (Discrete uniform PIT) *Let $Z_1, \dots, Z_n, Z_{n+1}$ be exchangeable observations with no ties almost surely. Then*

$$F_n(Z_{n+1}) \sim \text{Unif}\{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\}.$$

Indeed, the rank R of Z_{n+1} in the augmented sample $(Z_1, \dots, Z_n, Z_{n+1})$ is uniform on $\{1, \dots, n+1\}$. The augmented empirical CDF satisfies

$$F_{n+1}^{Z_{n+1}}(Z_{n+1}) = \frac{n}{n+1} F_n(Z_{n+1}) + \frac{1}{n+1} = \frac{R}{n+1},$$

so $F_n(Z_{n+1}) = (R-1)/n$. Thus its law is exact and distribution-free, but discrete on $n+1$ atoms. Unlike DKW, which controls F_n as an approximation to F , lemma 5 characterizes $F_n(Z_{n+1})$ directly. This rank-uniformity underlies the conformal framework introduced by Gammernan et al. (1998) and developed by Vovk et al. (2005) and Shafer and Vovk (2008).

3. Prediction sets on the empirical-CDF scale

Theorem 3 and lemma 5 now give two ways to construct a prediction set from F_n . We first build both sets on the same scale, then compare them.

The DKW prediction set. Fix a confidence level $1 - \delta$ and define the DKW radius

$$\varepsilon_n(\delta) = \sqrt{\frac{\log(2/\delta)}{2n}}.$$

Theorem 3 with $\varepsilon = \varepsilon_n(\delta)$ says with probability at least $1 - \delta$ over the calibration sample,

$$\sup_t |F_n(t) - F(t)| \leq \varepsilon_n(\delta). \quad (3)$$

Call the event in eq. (3) the DKW good event. On this event, $|F(z) - F_n(z)| \leq \varepsilon_n$ for every z simultaneously. This means that

$$F(z) \in [\alpha/2, 1 - \alpha/2] \implies F_n(z) \in [\alpha/2 - \varepsilon_n, 1 - \alpha/2 + \varepsilon_n].$$

The oracle prediction set is therefore contained in the inflated F_n -scale set whenever the DKW good event holds. We define the central DKW interval on the F_n -scale by

$$I_{\alpha, \delta, n}^{\text{DKW}} = [r_{\text{DKW}}, 1 - r_{\text{DKW}}] \cap [0, 1],$$

where $r_{\text{DKW}} = (\frac{\alpha}{2} - \varepsilon_n(\delta))_+$ and $(x)_+ = \max\{x, 0\}$. The DKW prediction set is

$$\hat{Q}_{\alpha, \delta}^{\text{DKW}} = \{z : F_n(z) \in I_{\alpha, \delta, n}^{\text{DKW}}\}.$$

The widening of the oracle target interval from $[\alpha/2, 1 - \alpha/2]$ to $[\alpha/2 - \varepsilon_n, 1 - \alpha/2 + \varepsilon_n]$ is the cost of substituting F_n for F . By construction, on the DKW good event, the conditional

coverage of $\widehat{Q}_{\alpha,\delta}^{\text{DKW}}$ given $\mathcal{D}_n$ is at least $1 - \alpha$. The DKW good event has probability at least $1 - \delta$. Therefore

$$\mathbb{P}_{\mathcal{D}_n} \left[\mathbb{P}(Z_{n+1} \in \widehat{Q}_{\alpha,\delta}^{\text{DKW}} \mid \mathcal{D}_n) \geq 1 - \alpha \right] \geq 1 - \delta. \quad (4)$$

This is a PAC guarantee. With probability at least $1 - \delta$ over the random calibration sample, the prediction set produced by DKW has conditional coverage at least $1 - \alpha$.

The DKW interval is non-trivial (not equal to all of $\mathbb{R}$) only when $r_{\text{DKW}} > 0$, that is, when $\varepsilon_n(\delta) < \alpha/2$. Solving gives the threshold

$$n > \frac{2 \log(2/\delta)}{\alpha^2}.$$

Below this threshold, DKW returns the trivial interval $[0, 1]$ on the F_n -scale, that is, all of $\mathbb{R}$. For $\alpha = 0.05$ and $\delta = 0.05$ the threshold is $n \approx 2951$. As $n \rightarrow \infty$ with δ fixed, $\varepsilon_n(\delta) \rightarrow 0$ and $r_{\text{DKW}} \rightarrow \alpha/2$. So the DKW interval converges to the oracle interval $[\alpha/2, 1 - \alpha/2]$, at rate $1/\sqrt{n}$.

The conformal prediction set. The conformal prediction set is built from lemma 5 the same way the oracle set is built from the continuous PIT. We pick a region in $[0, 1]$ with mass at least $1 - \alpha$ under the discrete uniform distribution of $F_n(Z_{n+1})$.

The support has $n + 1$ atoms, each carrying mass $1/(n + 1)$. A union of m consecutive atoms has mass $m/(n + 1)$. To get at least $1 - \alpha$ we need $m \geq (n + 1)(1 - \alpha)$, so we keep $\lfloor (n + 1)(1 - \alpha) \rfloor$ atoms and drop $\lfloor (n + 1)\alpha \rfloor$. Splitting the dropped atoms symmetrically across the two tails, we drop $\lfloor (n + 1)\alpha/2 \rfloor$ atoms from the left and $\lfloor (n + 1)\alpha \rfloor - \lfloor (n + 1)\alpha/2 \rfloor$ atoms from the right. The two counts differ by at most one, and only when $\lfloor (n + 1)\alpha \rfloor$ is odd. Define

$$r_{\text{conf}}^- = \frac{1}{n} \lfloor (n + 1)\alpha/2 \rfloor, \quad r_{\text{conf}}^+ = \frac{1}{n} (\lfloor (n + 1)\alpha \rfloor - \lfloor (n + 1)\alpha/2 \rfloor).$$

Then the conformal interval on the F_n -scale is

$$I_{\alpha,n}^{\text{conf}} = [r_{\text{conf}}^-, 1 - r_{\text{conf}}^+],$$

the direct discrete analogue of the oracle interval $[\alpha/2, 1 - \alpha/2]$. The two cuts differ by at most $1/n$ because the two tail counts differ by at most one. Their sum is $\lfloor (n + 1)\alpha \rfloor/n$, which differs from α by less than $1/n$. Therefore each cut differs from $\alpha/2$ by less than $1/n$, and both converge to $\alpha/2$ at rate $1/n$. This is a statement about the grid discretization and the marginal coverage overshoot. It is distinct from the stochastic convergence of the random order-statistic endpoints to the population quantiles, which is typically of order $n^{-1/2}$ on the probability scale.

The conformal prediction set is

$$\widehat{Q}_{\alpha}^{\text{conf}} = \{z : F_n(z) \in I_{\alpha,n}^{\text{conf}}\}. \quad (5)$$

Proposition 6 (Marginal validity of conformal prediction) *Under the assumptions of lemma 5,*

$$\mathbb{P}(Z_{n+1} \in \widehat{Q}_{\alpha}^{\text{conf}}) = \frac{n + 1 - \lfloor (n + 1)\alpha \rfloor}{n + 1} \geq 1 - \alpha.$$

Proof By lemma 5, $F_n(Z_{n+1})$ is uniform on the $n+1$ atoms of $\{0, 1/n, \dots, 1\}$. The interval $I_{\alpha,n}^{\text{conf}}$ contains exactly $n+1 - \lfloor (n+1)\alpha \rfloor$ of these atoms by construction. The probability is the number of contained atoms divided by $n+1$. ■

Marginal coverage is exactly $(n+1 - \lfloor (n+1)\alpha \rfloor)/(n+1)$, overshooting $1-\alpha$ by at most $1/(n+1)$. For marginal coverage at a fixed level, CP has only the rank-discretization overshoot

$$0 \leq \frac{n+1 - \lfloor (n+1)\alpha \rfloor}{n+1} - (1-\alpha) < \frac{1}{n+1}.$$

Realized coverage and the Beta law. We have two notions of coverage so far. CP gives marginal coverage at level $1-\alpha$ for any n . DKW gives high-probability conditional coverage at level $1-\alpha$ with confidence $1-\delta$. Given a calibration sample $\mathcal{D}_n$ and a constructed prediction set $\hat{Q}$, the conditional probability

$$C = \mathbb{P}(Z_{n+1} \in \hat{Q} \mid \mathcal{D}_n)$$

is itself a random variable: it depends on which calibration sample we drew. We call this the realized coverage. It is what the practitioner's set actually delivers after one calibration sample has been drawn. The marginal-coverage statement averages C over calibration samples, whereas the PAC statement bounds its lower tail. Theorem 4 gives the exact distribution of C for prediction sets built from order statistics. We now apply it to the conformal interval. For the conformal set in eq. (5), the two tail cuts are r_{conf}^- and r_{conf}^+ . Together they discard

$$n(r_{\text{conf}}^- + r_{\text{conf}}^+) = \lfloor (n+1)\alpha \rfloor$$

atoms from the F_n -grid. Theorem 4 therefore shows that its realized coverage satisfies

$$C_{\alpha}^{\text{conf}} := \mathbb{P}(Z_{n+1} \in \hat{Q}_{\alpha}^{\text{conf}} \mid \mathcal{D}_n) \sim \text{Beta}(n+1 - \lfloor (n+1)\alpha \rfloor, \lfloor (n+1)\alpha \rfloor),$$

when $\lfloor (n+1)\alpha \rfloor > 0$; otherwise the coverage is identically one. When the two tail cuts are symmetric, this is eq. (2). The Beta law places marginal and PAC validity on the same random variable. Marginal validity asks for $\mathbb{E}[C_k] \geq 1-\alpha$; since $\mathbb{E}[C_k] = (n+1-2k)/(n+1)$, this holds when $k \leq \lfloor (n+1)\alpha/2 \rfloor$. PAC validity instead asks for $\mathbb{P}(C_k \geq 1-\alpha) \geq 1-\delta$, a lower-tail condition that requires the Beta law to concentrate. Figure 1 makes the gap precise: ordinary CP can fall below its nominal realized coverage on nearly half of calibration draws, even though the size of the shortfall shrinks with n .

Exact PAC calibration. To compare conformal prediction with DKW under the same PAC guarantee, we use the Beta law to recalibrate the rank cut.

Given $\alpha \in (0, 1/2]$ and $\delta \in (0, 1]$, define

$$k_{\alpha,\delta,n} = \max \left\{ k \in \{0, 1, \dots, \lfloor n/2 \rfloor\} : \mathbb{P}(\text{Beta}(n+1-2k, 2k) \geq 1-\alpha) \geq 1-\delta \right\}, \quad (6)$$

with the convention that when $k=0$, the interval is the whole line and its realized coverage is $C_0 = 1$ almost surely. Equivalently, one may regard the degenerate law as a point mass at 1. Hence $k=0$ always satisfies the condition. The exact PAC conformal interval on the F_n -scale is

$$I_{\alpha,\delta,n}^{\text{PAC}} = [k_{\alpha,\delta,n}/n, 1 - k_{\alpha,\delta,n}/n],$$

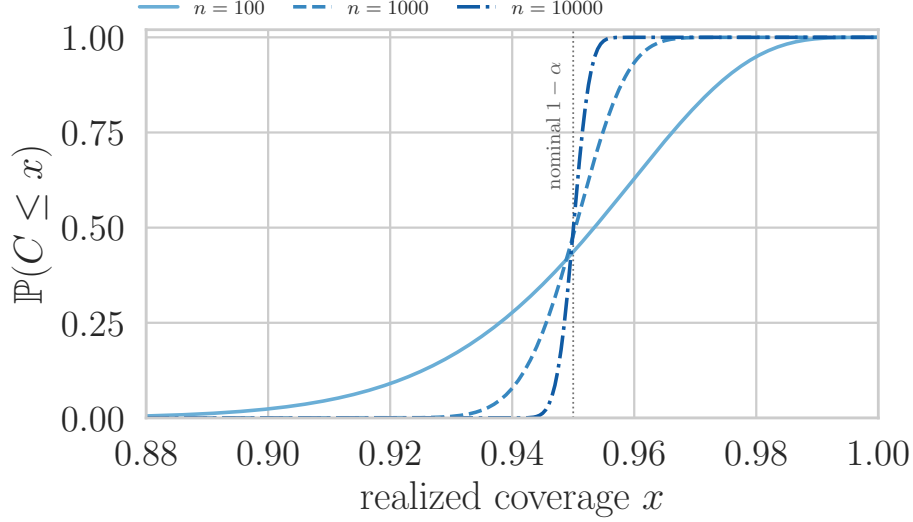


Figure 1: CDF of the realized coverage of ordinary CP at $\alpha = 0.05$. The vertical dotted line marks the nominal level $1 - \alpha = 0.95$. As n grows, the realized coverage concentrates around this level, while its probability of falling below it approaches one half.

and the corresponding prediction set is $\hat{Q}_{\alpha,\delta}^{\text{PAC}} = \{z : F_n(z) \in I_{\alpha,\delta,n}^{\text{PAC}}\}$.

By theorem 4 and the choice of $k_{\alpha,\delta,n}$, this set delivers

$$\mathbb{P}_{\mathcal{D}_n} \left[\mathbb{P}(Z_{n+1} \in \hat{Q}_{\alpha,\delta}^{\text{PAC}} \mid \mathcal{D}_n) \geq 1 - \alpha \right] \geq 1 - \delta,$$

exactly the same kind of guarantee as the DKW guarantee in eq. (4). The two methods are now in the same class.

Proposition 7 (Optimality on the symmetric rank grid) *Suppose F is continuous. For $0 \leq k \leq \lfloor n/2 \rfloor$, let*

$$\hat{Q}_k = \{z : F_n(z) \in [k/n, 1 - k/n]\}.$$

Among these deterministic symmetric rank-threshold sets, $\hat{Q}_{\alpha,\delta}^{\text{PAC}} = \hat{Q}_{k_{\alpha,\delta,n}}$ is the smallest by inclusion that is PAC-valid at level (α, δ) .

Proof By theorem 4, $\hat{Q}_k$ is PAC-valid exactly when k is admissible in eq. (6). Hence $k \leq k_{\alpha,\delta,n}$, and therefore $\hat{Q}_{\alpha,\delta}^{\text{PAC}} \subseteq \hat{Q}_k$. $\blacksquare$

This does not claim optimality over asymmetric cuts, randomization, different score functions, or problem-specific structure.

4. Comparing the prediction sets

All three prediction sets now live on the same F_n -scale. We first compare ordinary conformal prediction with DKW at the same nominal α . The DKW set is $\widehat{Q}_{\alpha,\delta}^{\text{DKW}} = \{z : F_n(z) \in I_{\alpha,\delta,n}^{\text{DKW}}\}$, and the conformal set is $\widehat{Q}_\alpha^{\text{conf}} = \{z : F_n(z) \in I_{\alpha,n}^{\text{conf}}\}$.

Proposition 8 (Ordinary CP contained in DKW) *For every $n \geq 1$, every $\alpha \in (0, 1)$, and every $\delta \in (0, 1]$,*

$$I_{\alpha,n}^{\text{conf}} \subseteq I_{\alpha,\delta,n}^{\text{DKW}}.$$

Proof Since $r_{\text{conf}}^+ \geq r_{\text{conf}}^-$, it is enough to show $r_{\text{conf}}^- \geq r_{\text{DKW}}$. For $n \geq 3$,

$$r_{\text{conf}}^- = \frac{1}{n} \left\lfloor \frac{(n+1)\alpha}{2} \right\rfloor > \frac{\alpha}{2} - \frac{1}{n}.$$

Moreover, $\varepsilon_n(\delta) \geq \sqrt{\log 2/(2n)} > 1/n$, so $r_{\text{conf}}^- > \alpha/2 - \varepsilon_n(\delta)$ and hence $r_{\text{conf}}^- \geq r_{\text{DKW}}$. For $n = 1$, $r_{\text{DKW}} = 0$. For $n = 2$, either $r_{\text{DKW}} = 0$, or $\alpha > \sqrt{\log 2} > 2/3$, in which case $r_{\text{conf}}^- = 1/2 > r_{\text{DKW}}$. $\blacksquare$

This first inclusion compares set sizes, not aligned guarantees. Ordinary CP gives marginal validity at level $1 - \alpha$, whereas DKW gives PAC validity at level $1 - \alpha$ with confidence $1 - \delta$. It therefore does not say that ordinary CP dominates DKW. The exact PAC conformal interval aligns the guarantees and gives the second comparison.

Proposition 9 (Exact PAC conformal contained in DKW) *Suppose F is continuous. For every $n \geq 1$, every $\alpha \in (0, 1/2]$, and every $\delta \in (0, 1]$,*

$$I_{\alpha,\delta,n}^{\text{PAC}} \subseteq I_{\alpha,\delta,n}^{\text{DKW}}.$$

Equivalently, $r_{\text{PAC}} := k_{\alpha,\delta,n}/n \geq r_{\text{DKW}}$.

Proof Set $k_* = \lceil nr_{\text{DKW}} \rceil$. Since F_n takes values on the grid $\{0, 1/n, \dots, 1\}$, the DKW interval is equivalent on that grid to $[k_*/n, 1 - k_*/n]$. By theorem 4, the corresponding order-statistic interval has realized coverage $\text{Beta}(n+1-2k_*, 2k_*)$. The DKW guarantee in eq. (4) says that this Beta variable is at least $1 - \alpha$ with probability at least $1 - \delta$, so k_* is admissible in eq. (6). Since $k_{\alpha,\delta,n}$ is the largest admissible rank,

$$r_{\text{PAC}} = \frac{k_{\alpha,\delta,n}}{n} \geq \frac{k_*}{n} \geq r_{\text{DKW}},$$

which gives the containment. $\blacksquare$

At the same PAC target (α, δ) , the exact PAC conformal interval is at least as small as the DKW interval on the F_n -scale, for every n and every F . The two methods deliver the same guarantee, but the conformal one is sharper because it computes the law of a single chosen order statistic (Beta) rather than the law of the supremum over all of them (DKW).

For the small- δ regime standard in PAC analysis, one has $I_{\alpha,n}^{\text{conf}} \subseteq I_{\alpha,\delta,n}^{\text{PAC}} \subseteq I_{\alpha,\delta,n}^{\text{DKW}}$. The first containment reflects that PAC validity is stronger than marginal validity, so the PAC procedure must produce a larger set. A larger set means a smaller tail cut: $r_{\text{PAC}} \leq r_{\text{conf}}^-$. This first containment can flip when δ is large, because the PAC requirement becomes weak and the PAC cut may exceed the marginal one. The robust universal statement is just $I_{\alpha,\delta,n}^{\text{PAC}} \subseteq I_{\alpha,\delta,n}^{\text{DKW}}$.

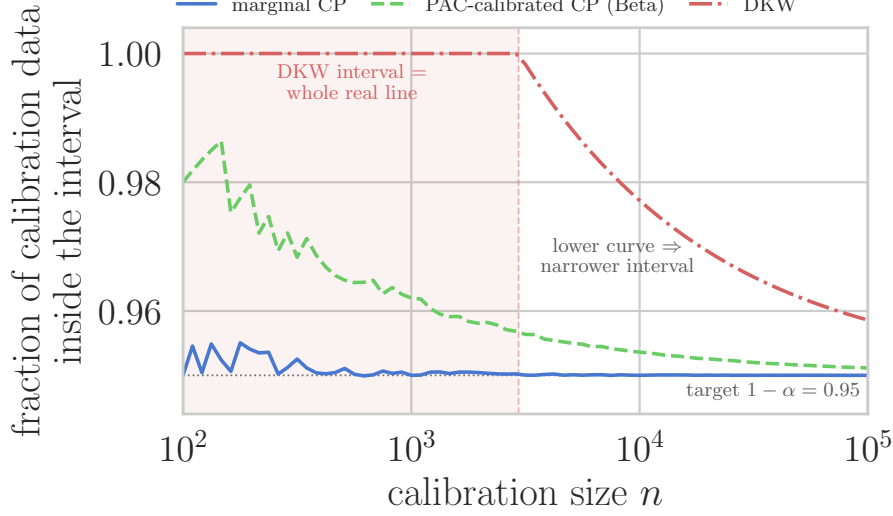


Figure 2: Fraction of the F_n -grid retained by the three intervals at $\alpha = \delta = 0.05$. Lower curves correspond to narrower intervals, and the oscillations in the CP curves come from the rank grid. In the shaded region, DKW retains the entire grid and its prediction set is the whole line.

5. Discussions

Let us consider few numerical illustrations. Figure 2 shows the fraction of the F_n -grid retained by each interval at $\alpha = \delta = 0.05$. A lower retained fraction means a narrower interval: marginal CP gives the smallest interval, exact PAC-CP an intermediate one, and DKW the largest. The exact tail cuts are reported in Appendix D. For a more stringent example, requiring the realized coverage of a one-sided interval to lie in $[0.94, 0.96]$ with probability 0.99999 needs $n_{\text{CP}}^* = 9,558$ under exact Beta calibration, compared with $n \geq 61,031$ when the two-sided DKW band is applied to that interval and $n \geq 244,122$ for a two-sided central interval. Appendix D gives the calculations. The preceding comparisons may suggest that DKW is uniformly worse than CP. That conclusion would be wrong. DKW is broader in scope: it supplies a single event covering all thresholds. The DKW guarantee is a single event:

$$\mathcal{E}_{\text{DKW}} = \left\{ \sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \leq \varepsilon_n(\delta) \right\}, \quad \mathbb{P}(\mathcal{E}_{\text{DKW}}) \geq 1 - \delta.$$

On this event the empirical CDF lies in a uniform band around F , simultaneously for every t . Any prediction-set construction that depends only on values of F_n inherits a coverage statement on $\mathcal{E}_{\text{DKW}}$. This is unlike the Beta law, which is a statement about one chosen order statistic.

CDF confidence bands and post-selection. For a confidence band around F , the DKW band $[F_n(t) - \varepsilon_n, F_n(t) + \varepsilon_n]$ is the canonical answer. The Beta law gives a tight pointwise interval at one chosen quantile. Combining many such intervals into a band requires a union bound and loses sharpness. Also A finite collection of Beta-calibrated intervals can be given joint validity by replacing δ with a Bonferroni allocation such as δ/m over m levels; this correction can remain substantially sharper than DKW for practical grids. DKW’s distinctive advantage is that one event covers the continuum of thresholds and levels, without specifying a grid or multiplicity correction in advance. It therefore remains valid when a level or threshold is selected after inspecting F_n , or when many downstream users choose different operating points from the same calibration sample. Some specially constructed data-dependent conformal procedures also retain validity, but not by the basic fixed-level theorem alone.

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Appendix A. Proof of Glivenko–Cantelli

Proof We follow the discretization argument of [Tsybakov \(2007\)](#). Fix $t \in \mathbb{R}$. The variable $F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{Z_i \leq t\}$ is the average of n *iid* Bernoulli($F(t)$) variables, with mean $F(t)$. By the strong law of large numbers,

$$F_n(t) \xrightarrow[n \rightarrow \infty]{\text{a.s.}} F(t), \quad \forall t \in \mathbb{R}. \quad (7)$$

To upgrade pointwise to uniform convergence, we discretize F . Fix an integer $k \geq 2$. Since F is continuous, there exist points $t_1 < t_2 < \dots < t_{k-1}$ such that $F(t_i) = i/k$ for $i = 1, \dots, k-1$. Set $t_0 = -\infty$ and $t_k = +\infty$ with the conventions $F(t_0) = 0$ and $F(t_k) = 1$. For any $t \in [t_{i-1}, t_i]$, monotonicity of F and F_n gives

$$\begin{aligned} F_n(t) - F(t) &\leq F_n(t_i) - F(t_{i-1}) = F_n(t_i) - F(t_i) + \frac{1}{k}, \\ F_n(t) - F(t) &\geq F_n(t_{i-1}) - F(t_i) = F_n(t_{i-1}) - F(t_{i-1}) - \frac{1}{k}. \end{aligned}$$

Taking absolute values and a maximum over i ,

$$\sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \leq \max_{1 \leq i \leq k-1} |F_n(t_i) - F(t_i)| + \frac{1}{k}. \quad (8)$$

By eq. (7) applied at the finitely many points $t_1, \dots, t_{k-1}$, the maximum on the right of eq. (8) converges almost surely to 0. Hence $\limsup_n \sup_t |F_n(t) - F(t)| \leq 1/k$ almost surely. Since k was arbitrary, sending $k \rightarrow \infty$ gives the result. $\blacksquare$

Appendix B. Massart’s first-crossing argument

The reduction in eq. (1) turns DKW into a statement about the uniform empirical process. Let $U_1, \dots, U_n$ be *iid* uniform on $[0, 1]$, let G_n be their empirical CDF, and write $U_{(1)} \leq \dots \leq U_{(n)}$ for their order statistics. Then

$$\sup_{u \in [0, 1]} (G_n(u) - u) = \max_{1 \leq j \leq n} \left(\frac{j}{n} - U_{(j)} \right).$$

Thus a one-sided deviation $\sup_u (G_n(u) - u) > \varepsilon$ occurs if and only if $j/n - U_{(j)} > \varepsilon$ for some rank j : DKW asks whether any order statistic crosses a slanted boundary.

For each rank, define the first-crossing event

$$A_j = \{j \text{ is the smallest index with } j/n - U_{(j)} > \varepsilon\}.$$

The events A_j are disjoint and partition the bad event. Hence

$$\mathbb{P}\left(\sup_u (G_n(u) - u) > \varepsilon\right) = \sum_j \mathbb{P}(A_j),$$

with no union-bound slack. The first-crossing formulas of [Smirnov \(1944\)](#) give closed-form expressions for the probabilities $\mathbb{P}(A_j)$, and Massart’s finite-sample argument bounds their

sum tightly enough to preserve the sharp exponential constant. For comparison, the one-sided maximum of a Brownian bridge B satisfies

$$\mathbb{P}\left(\sup_{u \in [0,1]} B(u) > \lambda\right) = e^{-2\lambda^2}.$$

The lower one-sided deviation is treated symmetrically, and combining the two bounds gives the factor 2 in theorem 3. Massart’s argument therefore controls whether any $U_{(j)}$ deviates too much from j/n . This uniformity supports confidence bands, post-selection guarantees, and risk-curve calibration. For one fixed prediction target, exact order-statistic laws can instead calibrate only the ranks that define the prediction set.

Appendix C. Finite future batches

For the one-sided interval $(-\infty, Z_{(b)}]$, theorem 4 gives the random realized coverage $C_b := F(Z_{(b)}) \sim \text{Beta}(b, n+1-b)$. Once the calibration sample has been drawn, the interval is fixed, and each further *iid* observation falls in it independently with probability C_b . Let $N_m := \sum_{j=1}^m \mathbb{1}\{Z_{n+j} \leq Z_{(b)}\}$ be the number covered among m future observations. The two layers of randomness give

$$N_m \mid C_b \sim \text{Bin}(m, C_b), \quad N_m \sim \text{BetaBinomial}(m, b, n+1-b).$$

The first law conditions on the chosen interval; the second averages over the calibration sample using Wilks’s Beta law.

As the future batch grows, the strong law gives $N_m/m \rightarrow C_b$ almost surely. Thus the binomial variation from the future observations disappears, but the calibration-sample variation remains: the limiting coverage is the random variable $C_b \sim \text{Beta}(b, n+1-b)$. Both the finite-batch law and its large-batch limit therefore follow directly from Wilks’s result; these formulations are studied in conformal language by Marques F. (2025).

Appendix D. Sample-size illustrations

For $\alpha = \delta = 0.05$, the triplets $(r_{\text{conf}}, r_{\text{PAC}}, r_{\text{DKW}})$ are $(0.0200, 0.0100, 0)$ at $n = 100$, $(0.0250, 0.0190, 0)$ at $n = 1000$, and $(0.0250, 0.0232, 0.0114)$ at $n = 10000$. The DKW cut vanishes when $n \leq 2 \log(2/\delta)/\alpha^2 \approx 2951$.

A more demanding target is $\mathbb{P}_{\mathcal{D}_n}(C \in [0.94, 0.96]) \geq 0.99999$. For the one-sided order-statistic interval underlying CP, set $K_n = \lceil 0.95(n+1) \rceil$. Under continuity of F , Wilks’s law gives $C_n = F(Z_{(K_n)}) \sim \text{Beta}(K_n, n+1-K_n)$. Exact evaluation of the Beta window probability shows that the smallest valid sample size is $n_{\text{CP}}^* = 9558$, with $K_n = 9082$.

Although the prediction interval is one-sided, the coverage window controls both directions of its error, so the two-sided DKW inequality is required. Its half-width is 0.01; for a two-sided central interval, the two endpoint errors require radius 0.005. Thus

$$n_{\text{DKW,one}} \geq \frac{\log(2/10^{-5})}{2(0.01)^2} = 61031, \quad n_{\text{DKW,two}} \geq \frac{\log(2/10^{-5})}{2(0.005)^2} = 244122.$$